SAT Math · Week 10 Practice Set

25 questions · 36 minutes

Name _____ Class _____ Date _____

1. No calculator

The functions f and h are defined by $f(x) = x^2 + 3x - 40$ and $h(x) = x^2$. If $3h(x) = f(2x)$, what is the positive value of x?

Student-produced response: enter your answer as a number.

2. No calculator

$$x^2(x+5)(x-b) = 0$$

In the given equation, b is a positive constant. The sum of the solutions of the equation is 7. What is the value of b?

Student-produced response: enter your answer as a number.

3. No calculator

If $\dfrac{x^{a^2}}{x^{b^2}} = x^{20}$, $x > 1$, and $a + b = 4$, what is the value of $a - b$?

Student-produced response: enter your answer as a number.

4. No calculator

a and b are positive consecutive integers, and $b > a$. If b is thirteen more than $\sqrt{a}$, what is the value of a?

Student-produced response: enter your answer as a number.

5. No calculator

If $x^2 - 8x + 5 = 0$ has roots r and s, what is the value of $r^2 + s^2$?

Student-produced response: enter your answer as a number.

6. Desmos

In the xy-plane, the graphs of $y = x^2 - 6x + 8$ and $y = 2x - 4$ intersect at two points. What is the sum of the x-coordinates of these two points?

Student-produced response: enter your answer as a number.

7. No calculator

If $s^2 = t^{-1}$ and $t^{1/3} = u^{-1/2}$, then what is the value of s in terms of u, for $s > 0$?

(A) $u^{-3/4}$
(B) $u^{-1/2}$
(C) $u^{1/2}$
(D) $u^{3/4}$

8. No calculator

The function N is defined by $N(t) = k \cdot 2^{-at}$ for positive constants k and a. The table shows values of N for select values of t.

t	0	1	2
$N(t)$	200	50	12.5

What is the value of a?

Student-produced response: enter your answer as a number.

9. No calculator

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of the circle?

(A) 3
(B) 4
(C) 5
(D) 25

10. No calculator

A circle in the xy-plane has the equation $x^2 + y^2 - 8x + 2y - 19 = 0$. What is the diameter of the circle?

Student-produced response: enter your answer as a number.

11. NO CALCULATOR

In the xy-plane, the graph of $x^2 + y^2 + 4x - 10y + k = 0$ is a circle with radius 3. What is the value of k?

Student-produced response: enter your answer as a number.

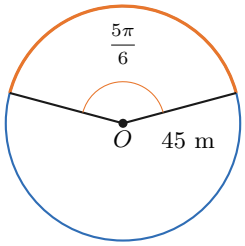
12. DESMOS

A circle has equation $(x + 4)^2 + (y - b)^2 = 100$, where b is a constant. The points $(4, 13)$ and $(-10, -1)$ both lie on the circle. What is the value of b?

Student-produced response: enter your answer as a number.

13. NO CALCULATOR

A circular plaza has a radius of 45 meters. A decorative walking path forms an arc along the plaza's boundary that subtends a central angle of $\frac{5\pi}{6}$ radians. What is the length of the arc, in meters?



(A) $\frac{75\pi}{2}$

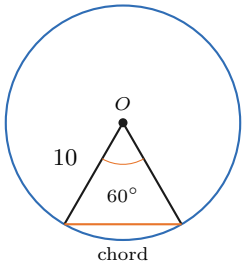
(B) 45π

(C) 150π

(D) $\frac{25\pi}{2}$

14. NO CALCULATOR

In a circle with radius 10, a chord subtends a central angle of 60°. What is the length of the chord?



(A) 5

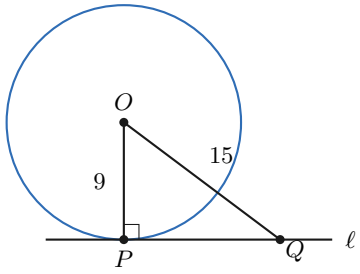
(B) $5\sqrt{3}$

(C) 10

(D) $10\sqrt{3}$

15. NO CALCULATOR

In the figure, line ℓ is tangent to the circle with center O at point P. If $OP = 9$ and $OQ = 15$ for a point Q on line ℓ, what is the length of PQ?



(A) 6

(B) 12

(C) $\sqrt{306}$

(D) 18

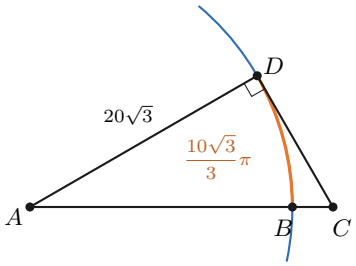
19. DESMOS

If $\sin(2x)° = \cos(x + 15)°$ and $0 < x < 45$, what is the value of x?

Student-produced response: enter your answer as a number.

16. CALCULATOR

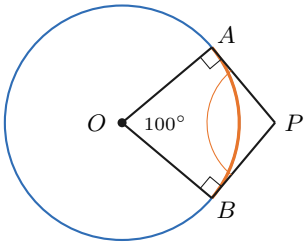
A circle has center A and radius $20\sqrt{3}$. Segment CD is tangent to the circle at point D, so triangle ACD has a right angle at D. Point B lies on the circle along ray AC, and minor arc BD has length $\dfrac{10\sqrt{3}}{3}\pi$. What is the length of CD?



Student-produced response: enter your answer as a number.

17. NO CALCULATOR

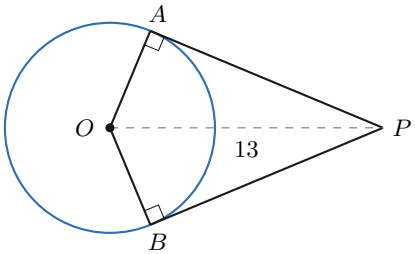
In a figure, PA and PB are tangent to a circle centered at O, at points A and B respectively, so $OA \perp PA$ and $OB \perp PB$. If $\angle APB = 100°$ and the circle has a circumference of 36π, what is the length of minor arc AB?



(A) 6π
(B) 8π
(C) 9π
(D) 10π

18. CALCULATOR

A circle has circumference 10π. Point P is 13 units from the center O of the circle. Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the circle at A and B. What is the area of quadrilateral $PAOB$?



Student-produced response: enter your answer as a number.

20. DESMOS

For $0 < x < 45$, the equation

$$\sin((5x - 20)°) = \cos((x + 14)°)$$

has more than one solution. What is the sum of all solutions x in this interval?

Student-produced response: enter your answer as a number.

21. No calculator

In right triangle PQR, $\angle R = 90°$ and $\sin P + \sin Q = \dfrac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

Student-produced response: enter your answer as a number.

22. Calculator

In a right triangle, angle θ is acute and $\sin\theta = 0.6$. What is the value of $\cos(90° - \theta) + \tan\theta$?

(A) 1.35
(B) 1.6
(C) 0.75
(D) 1.2

23. Calculator

A straight track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 0.3 kilometer long, measured along the slope. What is the vertical rise of the track, in meters?

(A) 180
(B) 240
(C) 225
(D) 400

24. Calculator

Maya walked from Noah's house 4 miles due north, then 3 miles due east, while Noah walked 2 miles due south and 5 miles due west. If Noah were to walk directly to Maya's location, how many hours would it take him, if Noah walks at a rate of 5 miles per hour?

Student-produced response: enter your answer as a number.

25. No calculator

If θ is an acute angle and $\tan\theta = \dfrac{9}{40}$, what is the value of $\cos\theta$?

(A) $\dfrac{40}{41}$
(B) $\dfrac{9}{41}$
(C) $\dfrac{40}{9}$
(D) $\dfrac{41}{40}$